

Supermart Analysis:

Sales, Customers & Product Insights

1. Project overview

This project analyzes retail data to understand sales performance, customer behavior, product trends, and return patterns. The data is cleaned using Excel and Power Query, explored using Python for EDA, analyzed using SQL, and visualized in Power BI to generate business insights.

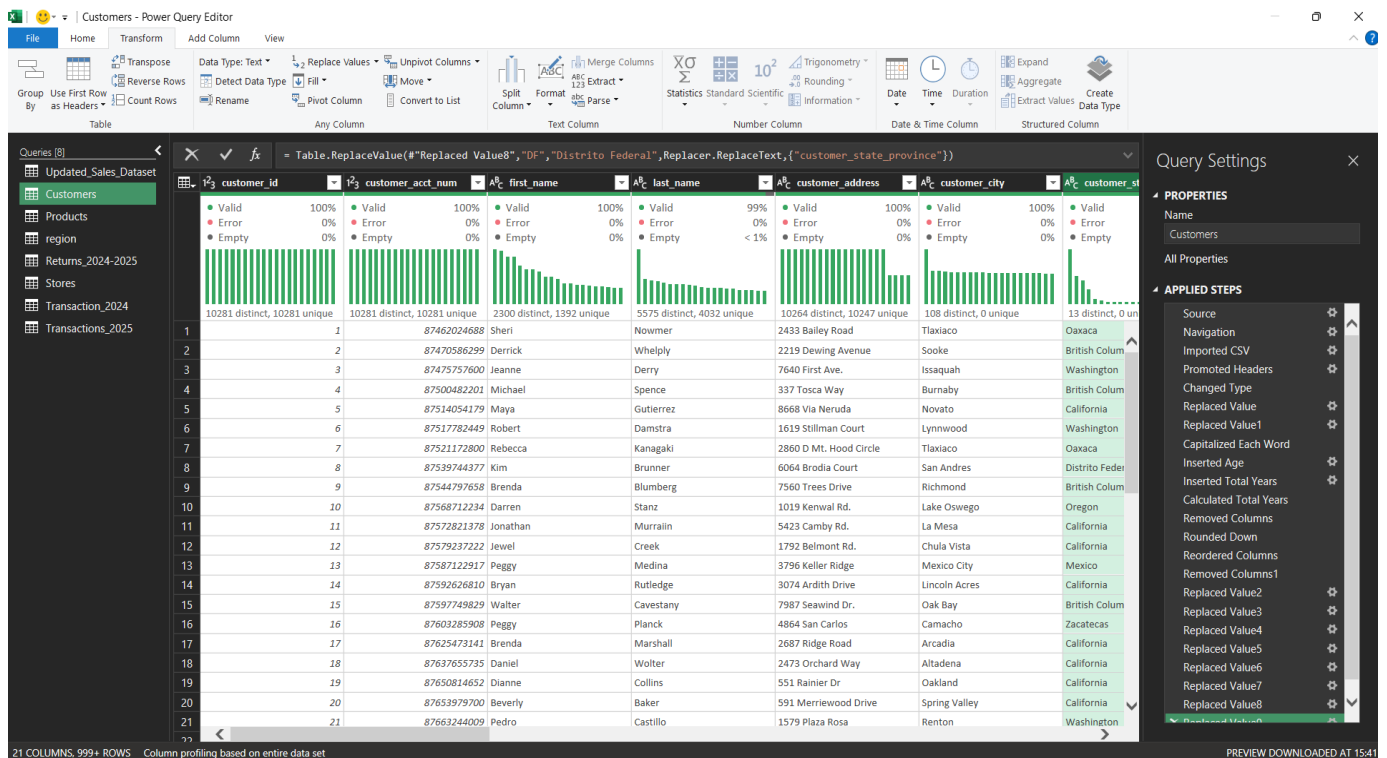
2. Dataset Summary

- The dataset includes five main tables: Customers, Products, Stores, Transactions (2024–2025), and Returns.
- These tables are connected using common keys such as customer ID, product ID, and store ID, allowing analysis across different aspects of the business.
- The transactions dataset serves as the core for sales and revenue analysis, while other tables provide supporting details for deeper insights.

3. Data Understanding and Cleaning

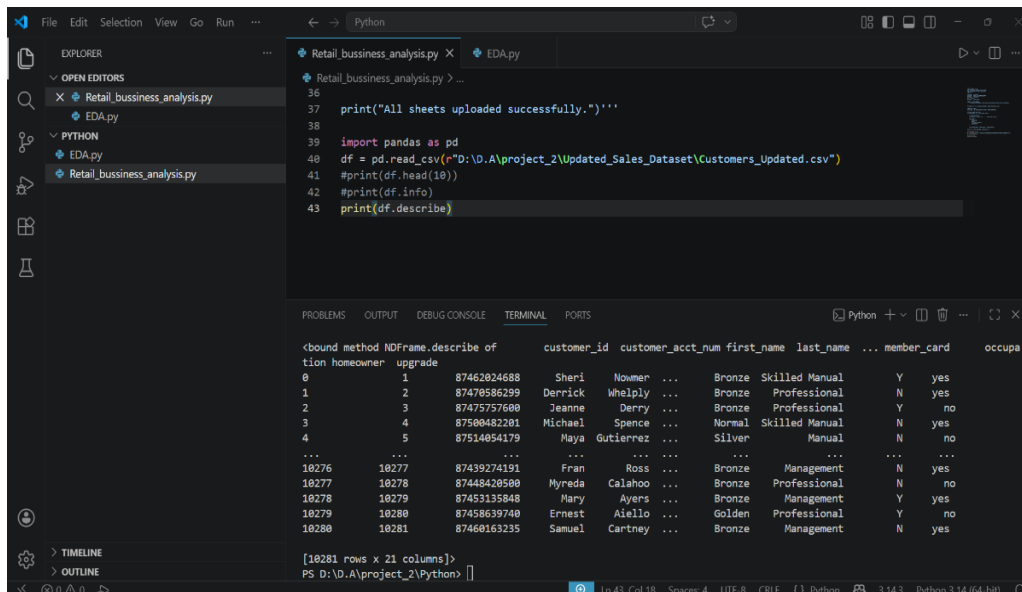
We began data preparation and cleaning in excel and power query

- **Data loading:** imported the dataset in powerquery (for cleaning and formatting)



4. Python role and EDA

- **Data Loading:** Imported the dataset to python using **pandas**.
- **Initial Exploration:** Used **df.info()** to check structure and **.describe()** for summary statistics.



The screenshot shows a Python IDE with a file explorer on the left and a terminal at the bottom. The main editor displays a script named `Retail_business_analysis.py`. The script imports `pandas` and reads a CSV file from a specified path. It then prints the first 10 rows of the DataFrame, its structure, and summary statistics. The terminal output shows the first 10 rows of the DataFrame, which includes columns for customer ID, account number, first name, last name, member card, and occupation.

```
36 print("All sheets uploaded successfully.")'''
37
38
39 import pandas as pd
40 df = pd.read_csv(r"D:\D.A\project_2\Updated_Sales_Dataset\Customers_Updated.csv")
41 #print(df.head(10))
42 #print(df.info)
43 print(df.describe)
```

	customer_id	customer_acct_num	first_name	last_name	...	member_card	occupa
0	1	87462024688	Sheri	Noumer	...	Bronze	Skilled Manual
1	2	87470586299	Derrick	Whelply	...	Bronze	Professional
2	3	87475757600	Jeanne	Derry	...	Bronze	Professional
3	4	87500482201	Michael	Spence	...	Normal	Skilled Manual
4	5	87514054179	Maya	Gutierrez	...	Silver	Manual
...	...	...	...	...	...	...	...
10276	10277	87430274191	Fran	Ross	...	Bronze	Management
10277	10278	87448420500	Myranda	Calahoo	...	Bronze	Professional
10278	10279	87453135048	Mary	Ayers	...	Bronze	Management
10279	10280	87458639740	Ernest	Aiello	...	Golden	Professional
10280	10281	87460163235	Samuel	Cartney	...	Bronze	Management

[10281 rows x 21 columns]
PS D:\D.A\project_2\Python>

- **Data Ingestion & Database Integration**
Automated the process of importing an 8-sheet Excel workbook into PostgreSQL using Python (**pandas** and **SQLAlchemy**).
- **Data Consolidation**
Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

```
1 import pandas as pd
2 from sqlalchemy import create_engine
3 from urllib.parse import quote_plus
4 import re
5
6 username = "postgres"
7 password = quote_plus("*****")
8 hostname = "localhost"
9 port = "5432"
10 database = "super_mart"
11
12 engine = create_engine(
13     f"postgresql+psycopg2://{username}:{password}@{hostname}:{port}/{database}"
14 )
15
16 file_path = r"D:\D.A\project_2\Retail Business Analysis.xlsx"
17
18 # Read all sheets
19 excel_data = pd.read_excel(file_path, sheet_name=None)
20
21 # Loop through each sheet
22 for sheet_name, df in excel_data.items():
23
24     # Clean table name
25     table_name = re.sub(r'W+', '_', sheet_name.lower())
26
27     df.to_sql(
28         table_name,
29         engine,
30         if_exists="replace",
31         index=False
32     )
33
34     print(f"Uploaded: {sheet_name} -> {table_name}")
35
36 print("All sheets uploaded successfully.")
37
```

5. Data analysis using SQL (Business Transaction)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue Drivers

Which products and categories contribute the most to total revenue?

→ Helps focus inventory and marketing

	product_id bigint	product_name text	product_brand text	total_revenue double precision
1	1428	Hermanos Green Pepper	Hermanos	2489.7000000000007
2	1380	Hilltop Mint Mouthwash	Hilltop	2447.12
3	362	Carlson Head Cheese	Carlson	2417.7300000000005
4	274	Great Pumpernickel Bread	Great	2408.56000000000036
5	941	Fabulous Strawberry Drink	Fabulous	2377.8999999999999
6	1467	Fort West BBQ Potato Chips	Fort West	2328
7	379	Moms Foot-Long Hot Dogs	Moms	2325.4
8	993	Even Better Large Curd Cottage Cheese	Even Better	2322.5999999999967
9	446	Red Wing Plastic Knives	Red Wing	2321.25
10	175	High Top New Potatos	High Top	2313.8499999999999
Total rows: 1559		Query complete 00:00:00.334		

2. Customer Value

Which customer segments generate the highest revenue? (income, education, etc.)

→ Helps target high-value customers

	income text	education text	total_revenue double precision
1	\$30K - \$50K	High School Deg...	364438
2	\$10K - \$30K	Partial High Sch...	354937
3	\$50K - \$70K	Bachelors Degree	231061
4	\$70K - \$90K	Bachelors Degree	130041
5	\$30K - \$50K	Partial College	117477
6	\$30K - \$50K	Partial High Sch...	77764
7	\$70K - \$90K	Graduate Degree	41015
8	\$130K - \$15...	High School Deg...	32573
9	\$70K - \$90K	High School Deg...	26278
10	\$50K - \$70K	High School Deg...	26198

3. Store Performance

Which stores perform best and worst in terms of revenue and transactions?

→ Identify expansion or closure decisions

	store_id bigint	store_name text	total_revenue double precision	transactions bigint
1	13	Store 13	170398.94000000008	25865
2	17	Store 17	157695.900000000052	23600
3	15	Store 15	115981.459999999979	17447

	store_id bigint 🔒	store_name text 🔒	total_revenue double precision 🔒	transactions bigint 🔒
22	2	Store 2	9519.130000000001	2769
23	14	Store 14	9324.939999999997	2766
24	5	Store 5	4903.200000000008	1470

4. Time Trends

How does sales vary over time (monthly / yearly)? Are there seasonal patterns?

→ Helps forecasting and planning

	month timestamp without time zone 🔒	total_revenue numeric 🔒
1	2024-01-01 00:00:00	45539.69
2	2024-02-01 00:00:00	44058.79
3	2024-03-01 00:00:00	50029.87
4	2024-04-01 00:00:00	42878.25
5	2024-05-01 00:00:00	44451.19
6	2024-06-01 00:00:00	45331.73
7	2024-07-01 00:00:00	50246.88
8	2024-08-01 00:00:00	46199.04
9	2024-09-01 00:00:00	43825.97
10	2024-10-01 00:00:00	42342.27
11	2024-11-01 00:00:00	53363.71
12	2024-12-01 00:00:00	56965.64
13	2025-01-01 00:00:00	98155.28
14	2025-02-01 00:00:00	94498.00
15	2025-03-01 00:00:00	98219.90
16	2025-04-01 00:00:00	94770.92
17	2025-05-01 00:00:00	95178.81
18	2025-06-01 00:00:00	97034.65
19	2025-07-01 00:00:00	98943.44
20	2025-08-01 00:00:00	94718.51
21	2025-09-01 00:00:00	101373.23
22	2025-10-01 00:00:00	92428.07
23	2025-11-01 00:00:00	113780.19
24	2025-12-01 00:00:00	120160.84

(Yes, revenue have seasonal patterns in the end month of year the revenue rise up)

5. Store-level Returns

Which stores have unusually high return rates?

→ Could mean staff issues, fraud, or poor inventory

	store_id bigint 🔒	total_sold numeric 🔒	total_returned numeric 🔒	return_rate numeric 🔒
1	8	40989	480	0.01171045890360828515
2	20	10600	121	0.01141509433962264151
3	23	22377	250	0.01117218572641551593
4	12	41800	460	0.01100478468899521531
5	16	52486	559	0.01065045917006439813

6. Return Analysis

Which products have the highest return rates?

→ Signals quality or expectation issues

	product_id bigint	product_name text	product_brand text	total_sold numeric	total_returned numeric	return_rate numeric
1	574	Shady Lake Spaghetti	Shady Lake	478	14	0.0293
2	1432	Hermanos Red Pepper	Hermanos	601	17	0.0283
3	581	Walrus Merlot Wine	Walrus	538	15	0.0279
4	564	Atomic Tasty Candy Bar	Atomic	467	13	0.0278
5	1461	Hermanos Oranges	Hermanos	554	15	0.0271
6	522	Tell Tale Walnuts	Tell Tale	534	14	0.0262
7	81	Blue Medal Egg Substit...	Blue Medal	496	13	0.0262
8	497	Tell Tale Cauliflower	Tell Tale	572	14	0.0245
9	833	Ebony Walnuts	Ebony	550	13	0.0236
10	1179	Horatio Sesame Crackers	Horatio	555	13	0.0234

7. Customer Behavior

Do repeat customers spend more than new customers?

→ Helps retention strategy

	customer_type text	total_customers bigint	total_revenue double precision	avg_spend_per_customer numeric
1	Repeat	7617	1735652.549999826	227.87
2	New	1225	28842.31999999894	23.54

Out of 10,281 registered customers, only 8,842 (~86%) have made purchases, while ~14% remain inactive."

8. Product Performance vs Returns

Are high-selling products also high-return products?

→ Classic business trade-off

	product_id bigint	total_sold numeric	total_returned numeric	return_rate numeric
1	502	698	5	0.00716332378223495702
2	426	680	9	0.01323529411764705882
3	1380	676	6	0.00887573964497041420
4	382	675	4	0.00592592592592592593
5	1362	665	12	0.01804511278195488722
6	277	663	9	0.01357466063348416290
7	440	658	5	0.00759878419452887538
8	827	653	2	0.00306278713629402757
9	1409	651	8	0.01228878648233486943
10	813	648	5	0.00771604938271604938

(high-selling product also high-return product often incorrect)

9. Demographics vs Spending

How do age, income, or occupation affect purchase behavior?

→ Helps segmentation and personalization

	age_group text	avg_spend numeric	avg_frequency numeric
1	Middle Age	201.55	6.66
2	Senior	200.49	6.61
3	Young	192.23	6.39

10. Sales Efficiency

Which stores generate the highest revenue per customer?

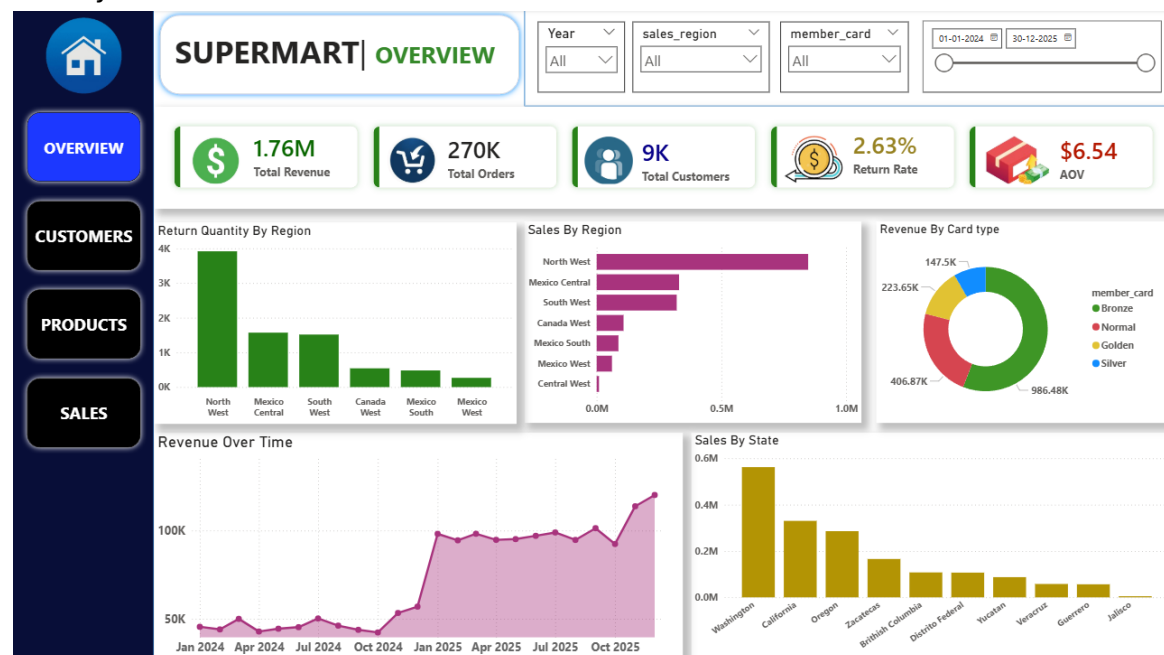
→ Measures operational efficiency

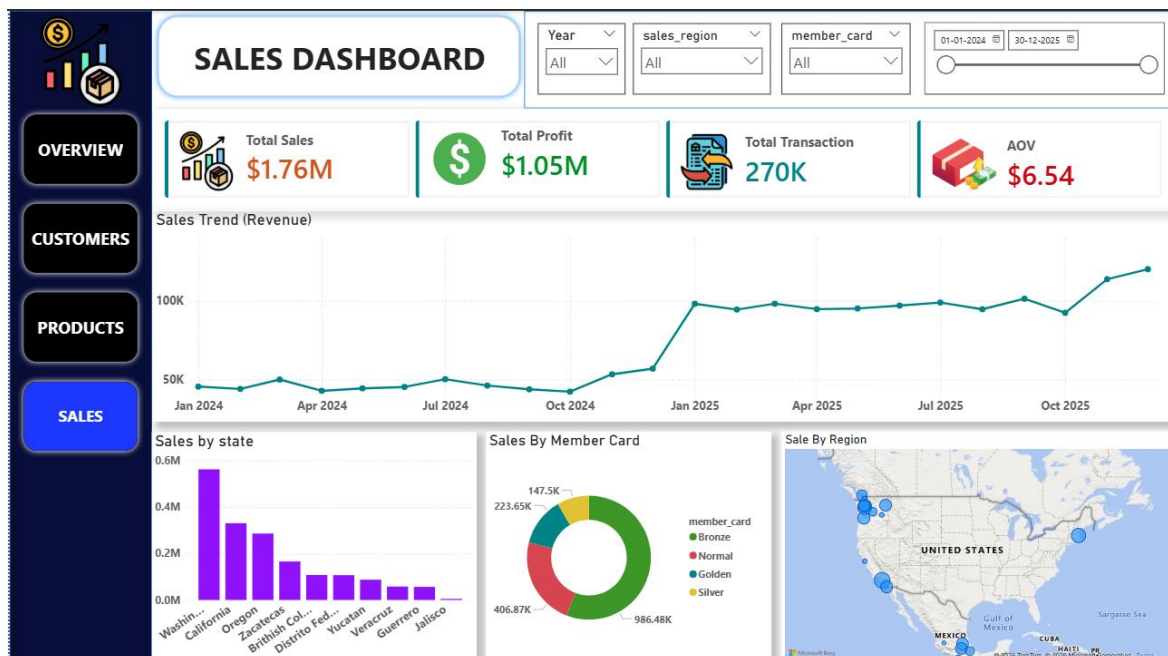
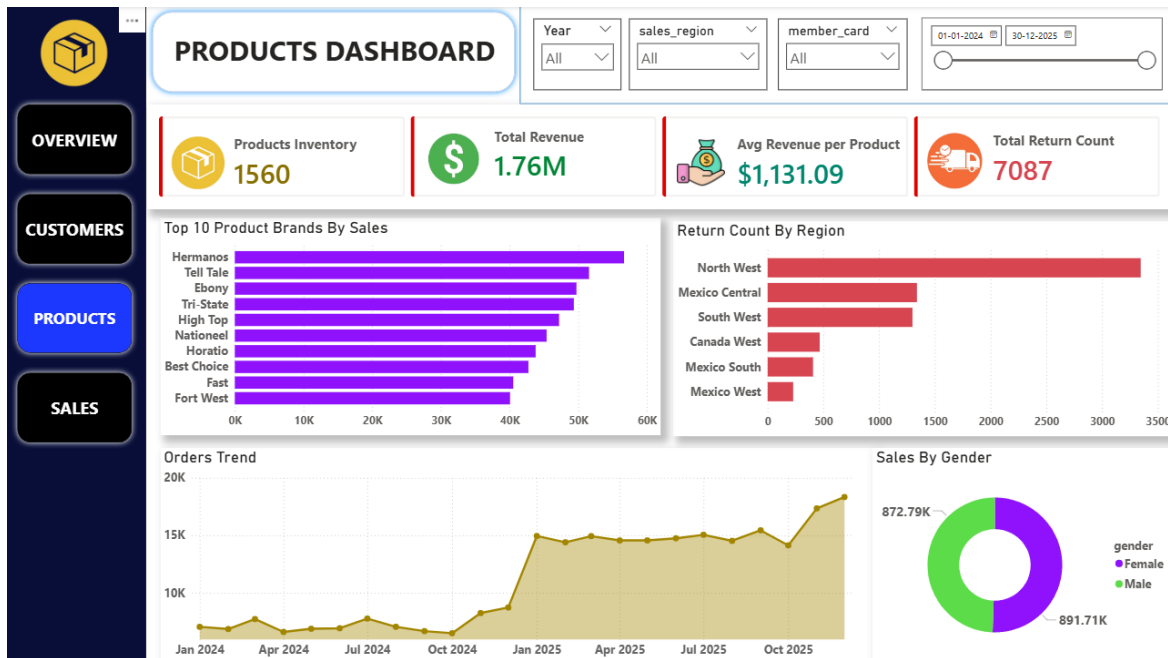
	store_id bigint	store_name text	total_revenue double precision	total_customers bigint	revenue_per_customer numeric
1	16	Store 16	110480.060000000055	84	1315.24
2	8	Store 8	87240.590000000004	98	890.21
3	12	Store 12	87613.360000000035	99	884.98
4	10	Store 10	57835.359999999993	92	628.65
5	3	Store 3	110877.880000000048	179	619.43

Out of 24 stores these are the top 5 stores generates highest revenue per customer

6. Dashboard in Power BI

Finally, we built an multi-page interactive dashboard in **Power BI** to present insights visually.





7. Business Recommendations

- Double down on high-performing products**
 Increase inventory and visibility of top revenue-generating products
- Activate the 14% inactive customers**
 Run re-engagement campaigns (discounts, email nudges, reminders)
- Focus on high-value customer segments**
 Identify segments contributing most revenue (income, behavior, etc.)

- **Optimize store performance**

Study top-performing stores and replicate their strategies

- **Use seasonal trends for planning**

increase inventory and marketing before peak periods (year-end spike)

- **Improve revenue per customer**

Upsell and cross-sell strategies, Personalized recommendations, Optimize pricing strategies per segment